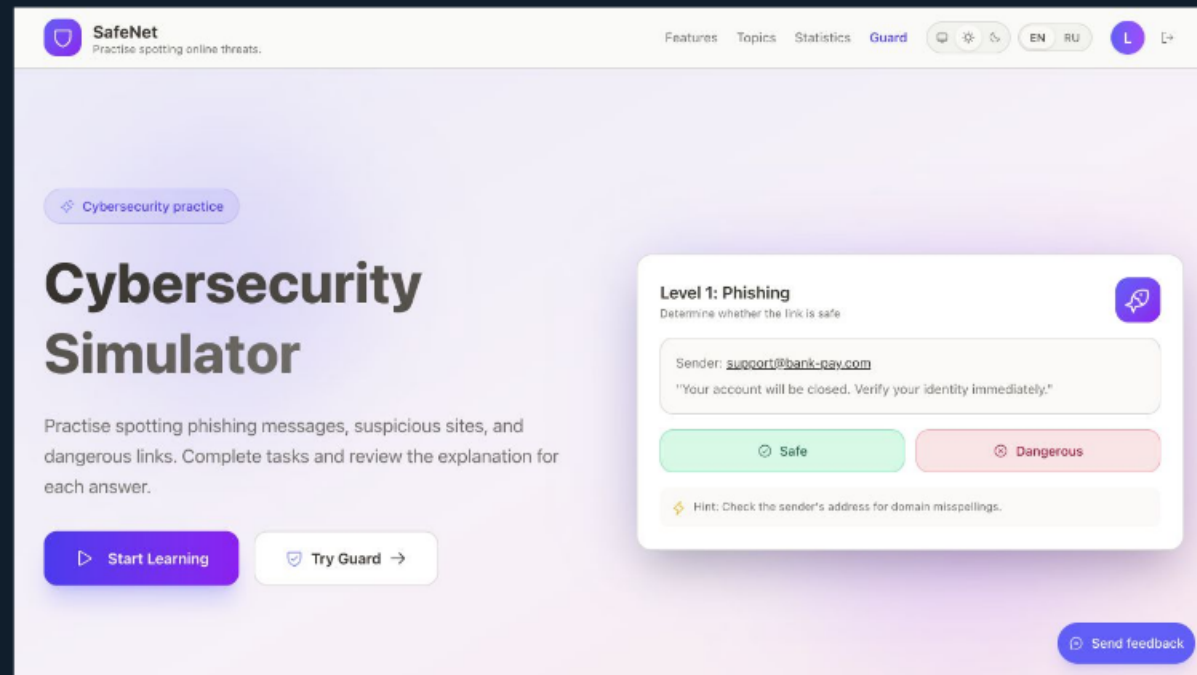


Safe-Net

Cybersecurity practice.
URL warnings explained.



Volkov Artem Aleksandrovich

Learning web app · Guard extension · Local prototype

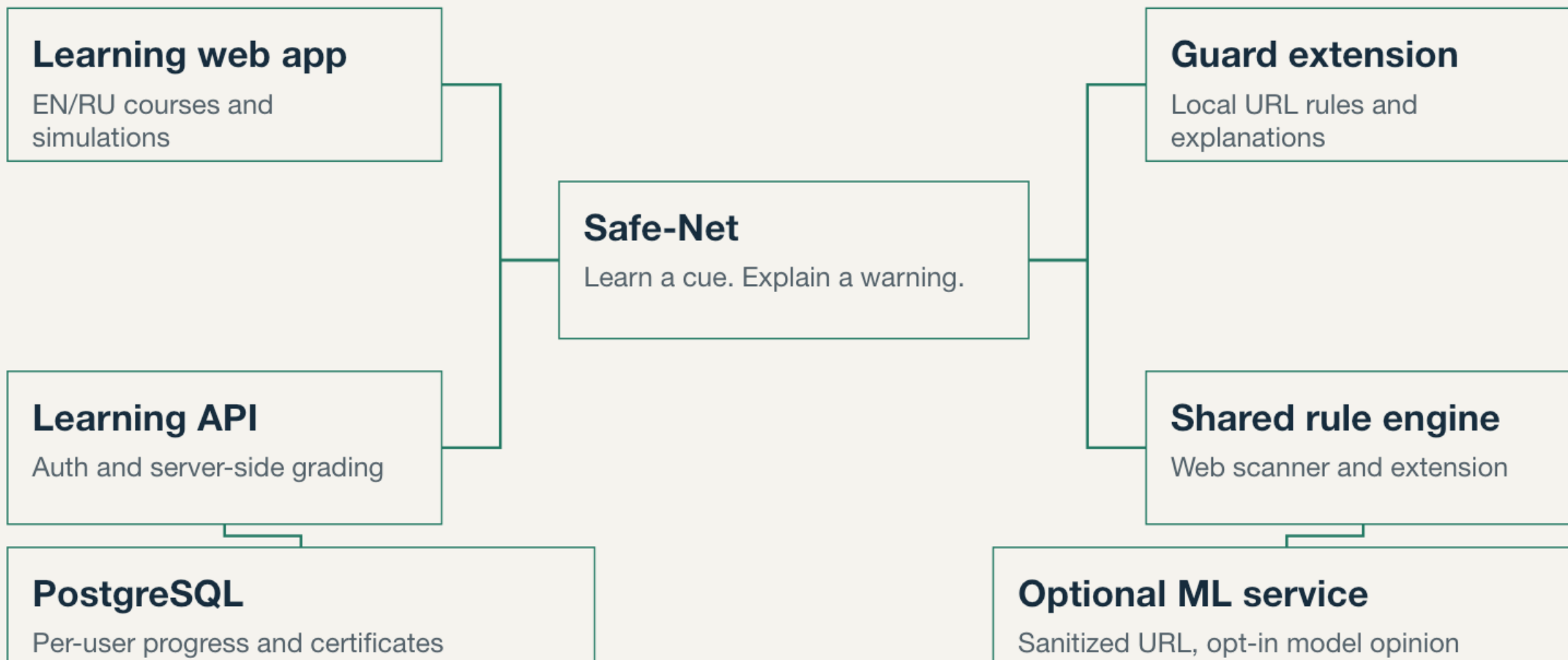
The research question

Can practice with explanations improve recognition of unfamiliar phishing cues?

Proposed comparison

Interactive practice versus an equal-duration text lesson.
Assess new examples immediately and after a delay.

How the system works



Inside the repository

client/

Next.js web app: courses, tasks and URL scanner

server/

NestJS API: accounts, grading and saved progress

extension/

WXT / Manifest V3: navigation and page checks

packages/guard-core/

TypeScript URL rules shared by web and extension

ml-service/

Optional FastAPI service and third-party BERT

server/content/

Versioned lessons and tasks, validated before seeding

The learning experience

21 courses

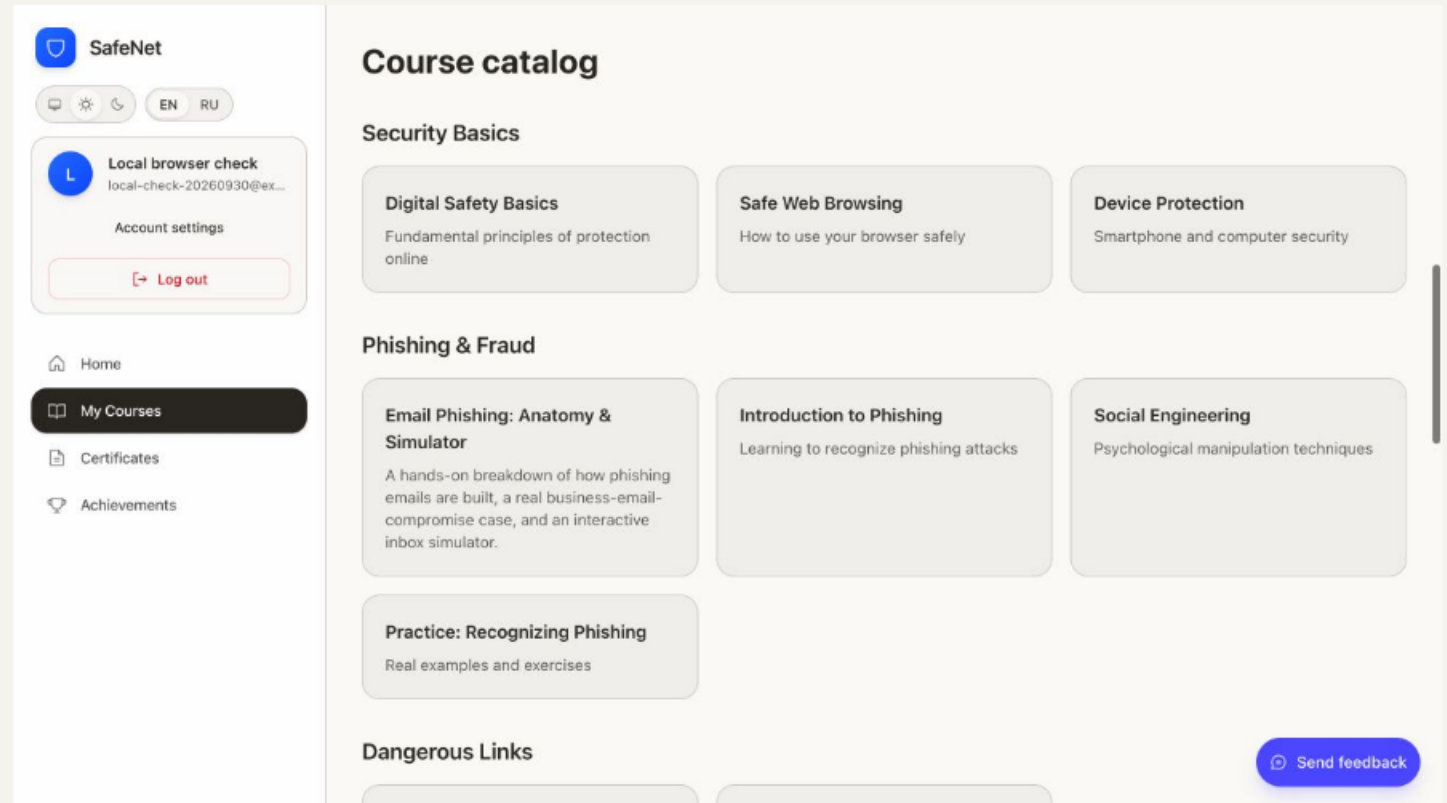
27 lessons

163 tasks

21 final tests

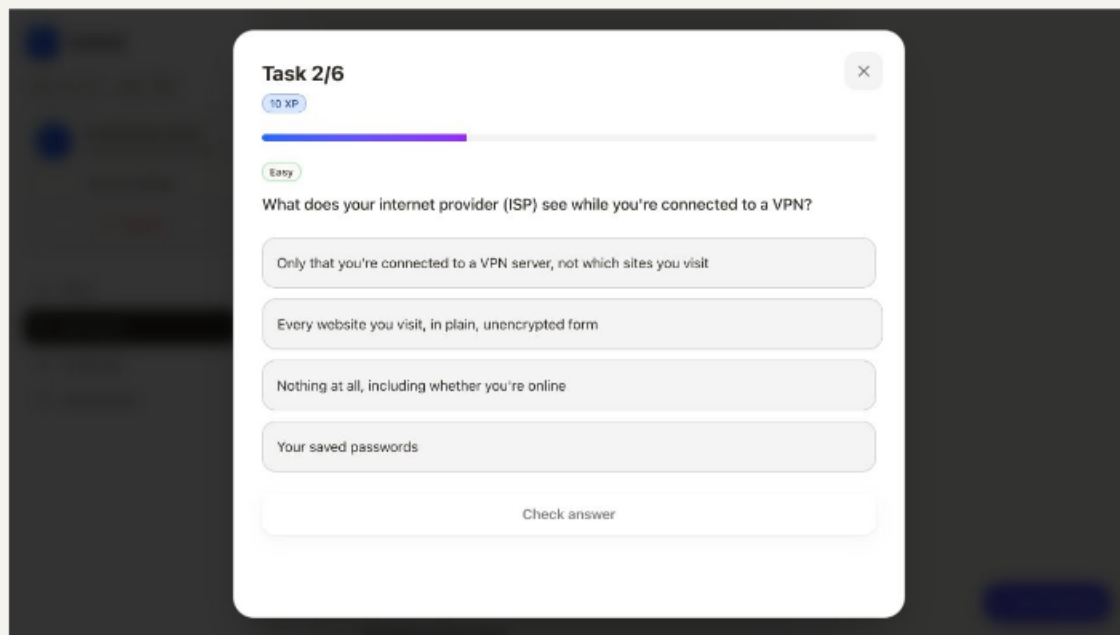
English and Russian

Lessons, practice tasks,
progress and certificates.



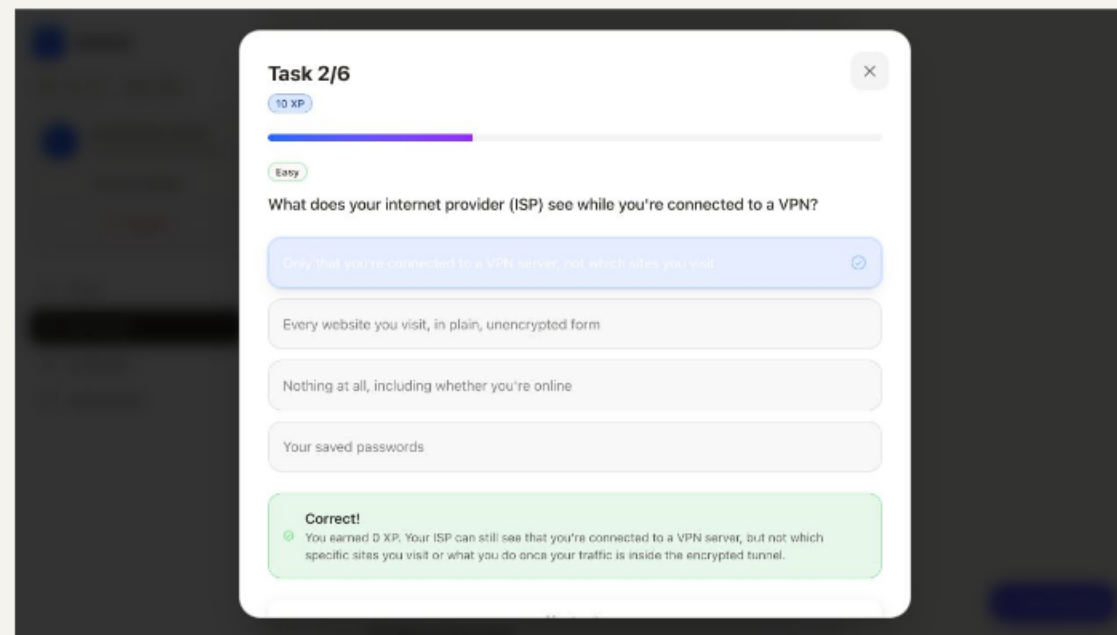
Local demo account · 30 September 2026

Practice, then an explanation



Answer a concrete question

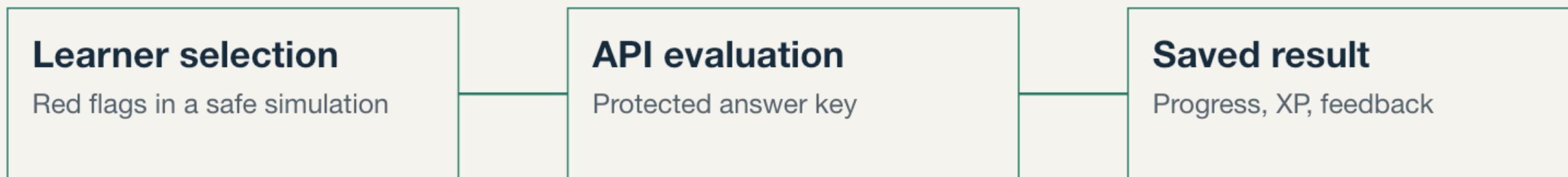
Choose what an ISP can see when traffic goes through a VPN.



Read the reason

Feedback explains the answer. Repeating a task adds no XP.

Server-side assessment



A completed task belongs to the learner account

Database constraints prevent repeated XP and duplicate course certificates.
A fresh tab reloads progress from the server.

Guard: the reasons behind a warning

88 / 100

Dangerous

Brand spelling changed
“o” replaced by zero

Suspicious URL wording
“login” in the path

All three run `@safe-net/guard-core` — one implementation. Earlier, separate rule sets produced different results: the courses used `paypal.com` as a phishing example while the detector scored it 8/100, safe. The shared engine now returns the same result for this example, and a regression test checks it.

Try it

Paste an address to run the checks in your browser.

`https://micros0ft-alerts.com/login`

Scan →

`sberbank.ru/login` Cyrillic “a”

`paypal.com` Greek “a”

`paypal.com` digit 1 for l

`micros0ft-alerts.com` zero for o

`sberbank.com.verify-account.info/login` brand as subdomain

`mail.google.com` legitimate domain

LOCAL VERDICT

88 / 100 **Dangerous**

- “microsoft” is embedded in a longer domain with letter-to-number substitutions.
- The URL contains a potentially manipulative word: “login”.

Send feedback

A rule score is a warning policy, not a probability of phishing.

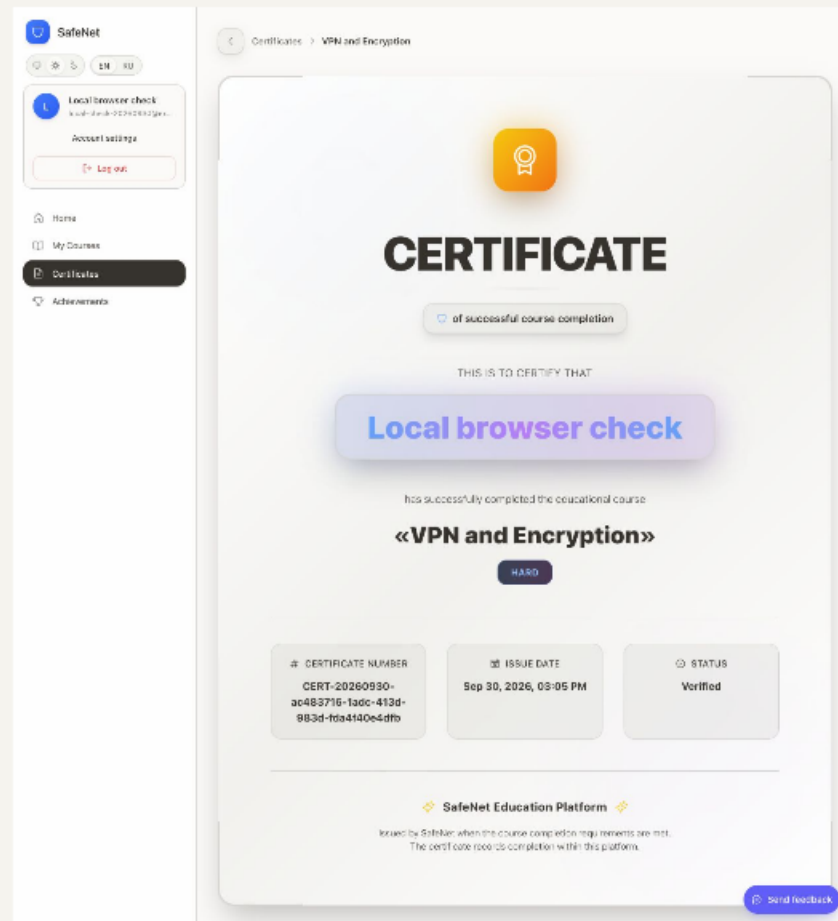
Progress and course completion

Saved per account

Completed tasks and tests
Course progress and XP
One certificate per course

Demo completion

VPN and Encryption
6 tasks · 8 test questions
95 XP · 1 certificate



This is a platform completion record.

Privacy and model boundaries

Default: local rules

The browser can score URLs without sending them to the ML service.

Optional: model opinion

Explicit opt-in sends a sanitized URL to the configured endpoint.

What remains in the optional request

Hostnames and paths still reveal information.
The BERT checkpoint comes from a third party.
Independent model evaluation remains open.

Engineering evidence

98

API regression tests

30

URLs in the rule parity corpus

Local builds and package checks passed
HTTP checks covered auth, grading and concurrency
Mailpit received EN/RU account emails

Current scope and remaining checks

Research outcomes

- No measured learning gains
- No independent detection accuracy
- No user study yet

Release acceptance

- Chrome installation and behavior
- External inbox delivery and hosted CI
- Accessibility and broader browser coverage

Current scope: a working local prototype

Suitable for technical feedback and study-design review.

Proposed learning-transfer study



Primary outcome

Change in balanced accuracy on delayed, unfamiliar examples

Also report phishing recall, benign false alarms and attrition.
Separate item banks reduce memorization of repeated examples.

Next step: evaluate the detector

Detector experiment

Build a versioned, independently labelled URL set with source-separated splits.

Error analysis

Compare rules, optional model and blend.
Report false positives by attack family and language.

Deliverable

An evaluation script, documented dataset procedure and a short report that explains failure cases.

Feedback requested

How should I distinguish learning transfer from memorizing familiar phishing cues?

Which assessment design would make a small pilot credible?

Would error analysis or a learning study be the more useful next step?